Shunchi Zhou

480.787.9252 • szhou82@asu.edu• linkedin.com/in/shunchi-zhou/• shunchizhou.com

SUMMARY

B.S.E. software engineering, pursuing a master's degree in computer science at Ira A. Fulton Schools of Engineering, Arizona State University. A passionate software developer with experience in back-end server development, data mining, machine learning, distributed database systems, and android application development.

EDUCATION

M.C.S., Computer Science (Big Data Systems) Aug. 2018 - May. 2020
Arizona State University, Tempe, AZ

B.S.E., Software Engineering Sept. 2013 - Jun.2017
Nantong University, China

PROJECTS EXPERIENCE

T-LeadLine: Interactive Visual Analysis of Textual Data (Topic Discovery & Event Exploration) 2018.08 - 2018.12

- A textual data driven event identification and exploration system
- Topic model building, events extracting and summarizing with the format of abstraction, spatial-temporal information and related person network, extended with topic ranking feature
- Major techniques including Named-entity Recognition (NER), Latent Dirichlet Allocation (LDA), detecting topic burst change (CUSUM), finding time-sensitive terms of given topic and time (TFIDF) and visualizing results with D3.js

Human Activity Recognition from Myo sensors 2018.08 - 2018.12

- A computing system that can understand human activities. a) identify known activities, b) segment sequence of activities, and c) identify unknown activities.
- Data collection: wore Myo wristband to record activities with timestamps. Feature extraction: extracted statistical features (min, max, avg, std, RMS). Dimension reduction: Principal Component Analysis (PCA), Singular Value Decomposition (SVD)
- Trained Decision Tree Classifier (DTS), Support Vector Machines (SVM), and Artificial Neural Networks (ANN) with the training data and then use the test data to report accuracy

Geo-spatial Hotspot Analysis based on Hadoop and Spark Using Scala language 2019.01 - 2019.05

- Deployed Apache Spark in a Hadoop cluster on the AWS, implemented functions to do spatial queries and hot spot analysis on the dataset of New York City Yellow Cab Taxi Trip Data
- Hot zone analysis: performed a range join operation on a rectangle dataset and a point dataset to obtain the number of points located within the rectangle; calculated the hotness of all the rectangles
- Hot cell analysis: identified statistically significant spatial hot spots using Getis-Ord G_i^* statistic

Customized Weight Control Android Application 2019.08 - 2019.12

- An android application aimed to control weight by providing exercise suggestions based on current health information combined with the number of expected calories to burn and food consumption in a single day
- Used Android Studio as the IDE, JDK 8, SQLite database. Designed UI, wrote codes for different components
- Team leader, main contributors to the project

WORK EXPERIENCE

Arizona State University, Graduate Service Assistant (Part-time) 2019.08 - 2019.12

- Provided academic support for undergraduates in programming skills
- Assisted with communication between professor and students

HONORS

- Full Scholarships for Students Studying Abroad at the University of Manchester, UK
- Academic Scholarship for Undergraduates (top 10%)
- Group Award for Undergraduate Service Outsourcing Entrepreneur Competition

TECHNICAL SKILLS

Programming Skills: Java, Python (pandas, numpy, scikit-learn, matplotlab, flask, pytorch), C++, JavaScript, Scala
Database Systems: SQL, MySQL, SQLite, Hadoop, Spark, PostgreSQL
Others: D3.js (Data Visualization), AWS, Git, Shell Scripting, Android, MATLAB, Docker